2 Precautions

This chapter describes Safety Information, Environmental Conditions, and Utility Requirements.

General Precautions

WARNING

- **If you have a pacemaker or similar implanted device, consult your physician before using this instrument. This instrument can potentially affect pacemaker operation.**
- **The power cord set that is shipped with this instrument cannot be used with other products.**
- **Do not use any power cord other than that supplied with this instrument.**

WARNING

Do not edit or change instrument parameters with any software except MassHunter.
This may result in impairment of the safety designs equipped in the instrument.
Also take sufficient precautions if you customize MassHunter operations using script files or other means.

CAUTION

The Agilent ICP-MS is a very safe instrument with many built-in protective features:

- A safety interlock shuts off the plasma if either of the top covers is opened during operation.
- A plasma viewing window provides eye protection.
- Sensors monitor the water and argon flow rate and pressure. If the water flow is too low to sufficiently cool the ICP-MS, or if the argon supply is insufficient, the plasma is turned off automatically.
- There are fans to help ensure that the instrument's internal temperature stays within the preset limits. The plasma is turned off automatically if any of these fans, except the QP fan, fail.
- Temperature sensors on the exhaust vent, instrument inside, and intake water line stop the plasma if overheating occurs.
- A manual switch shuts down the instrument immediately, even if the MassHunter Workstation is inoperable.

CAUTION

In case of uncertainty about a specific fluid, that fluid should not be used until the manufacturer confirms that it does not present a Hazard.

CAUTION

The bundled power supply cable and foreline pump power supply cable are for use exclusively with the ICP-MS. Do not use them for other devices or instruments.

CAUTION

Dealing with liquid spills

Refer to and follow the safety instructions on the MSDS (Material Safety Data Sheet) when available. Keep the instrument and the area around it clean. Simple spills such as foreline pump oil may be cleaned with a dry cloth, to be disposed of according to local regulations. In the case of large spills, use of a spill kit is recommended.

CAUTION

Before you use any cleaning or decontamination methods that are not specified by Agilent, please check with Agilent to make sure that the proposed method will not damage the equipment.

NOTE

To shut down the instrument quickly, turn OFF the main switch that is on the lower center of the front panel. To switch the instrument to Standby mode and/or Shutdown mode without using MassHunter Workstation, a “**Vacuum ON/OFF Switch**” is available inside of the front panel.

CAUTION

- A sensor monitors the radio frequency (RF) generator and shuts it down if it is improperly matched to the RF coil.
- A fiber-optic sensor shuts off the RF high voltage power whenever the plasma is not present or is turned off manually.
  The ICP-MS MassHunter Workstation displays warning messages when the sensors first detect a problem, which let you stop work before the shutdown limits are reached.
- The parameters that are displayed in the instrument control window change from green to yellow/red as the shutdown limits are approached or reached.

In addition to the above safety features, the following precautions should always be taken during operation or maintenance:

- Check acid concentration. Continuous aspiration of a highly concentrated acid might damage the inside of the instrument. Refer to the “Acid and Alkaline” section in the MassHunter Workstation online Help for more detail.
- Close the instrument covers and panels to operate the instrument.
- Check the exhaust system for a positive extraction at the exhaust duct.
- Handle solvents correctly.
- Check the drain vessel frequently.
- Wait for the instrument to cool before you do any maintenance procedure.
- Check the condition of the pipes, and replace them as needed.

Be sure to follow the cautions and warnings within this manual and the MassHunter Workstation online Help.

Precautions for Use

Protective Earth

WARNING

Connecting the ICP-MS to a power source that is not equipped with a protective earth contact creates a shock hazard for the operator and can damage the ICP-MS. Likewise, interrupting the protective conductor of the ICP-MS or overriding the power cord ground creates a shock hazard for the operator and can damage the instrument.

Closing the Instrument Panels and Covers

Close the instrument top covers before you start the plasma. Because the instrument can be optimized entirely via MassHunter Workstation, there is no need to open the front and side panels or cover after the plasma is ignited. In addition, a safety interlock shuts off the plasma if the cover is lifted.

WARNING

The external covers of the instrument protect the operator from internal hazards while the instrument is in operation. Exposure to high voltage and radio frequency radiation from the RF power supply is present and can be very dangerous.

Wearing Safety Glasses

WARNING

Eye Hazard

Always wear appropriate safety glasses when handling sample solutions and other chemicals, or when the plasma is on, in order to minimize the risk of eye damage by hazardous liquids and exposure to ultraviolet rays.

Checking the Exhaust System

WARNING

Asphyxiation Hazard

Laboratory must be equipped with ventilation system to supply continuous and sufficient fresh air from outside to prevent user from asphyxiation.

WARNING

Health Hazard, Asphyxiation Hazard

Laboratory must be equipped with exhaust system to remove harmful gases via a scrubber system to outside of the laboratory, to prevent user from asphyxiation. User safety requires that the exhaust gases from the plasma and vacuum systems be vented externally to the building and not recirculated by the environmental control system. Health hazards include chemical toxicity of solvents, samples, and foreline pump fluid vapor.

WARNING

Health Hazard

If the exhaust system stops for any reason, immediately shut off all gas supplies at the source.

- **Do not turn off the exhaust system while gas supplies are on.**
- **Do not cover the holes for air intake on the ICP-MS.**
- **If the exhaust fan is inoperable or there is insufficient air flow, do not start the ICP-MS until the appropriate maintenance personnel have repaired the problem.**

NOTE

The exhaust system must remain on at all times while the ICP-MS is on: in Analysis mode, Standby mode and Shutdown mode.

WARNING

Asphyxiation Hazard

If the exhaust system stops for any reason when using the “Gas Purge at Standby” function, immediately shut off all gas supplies at the source. Asphyxiation due to excess argon in the air may occur if the exhaust system is turned off while gas supplies are on.

WARNING

Noxious Gases

If the exhaust system stops for any reason while the vacuum pump or ICP-MS are still running, immediately turn on the exhaust system or turn off the pump and instrument. If inadequate ventilation occurs, vaporized pump fluid, ozone, and other toxic combustion products will accumulate in the laboratory. Hydrofluoric acid (HF) fumes, if inhaled, cause extensive burning of lung tissue.

WARNING

Explosion Hazard

The laboratory exhaust system must always be operational and connected to the instrument exhaust vent. Potential health hazards and the risk of explosions may occur if exhaust gases are incorrectly ventilated.

WARNING

Personal Injury Hazard

Check that the hinge for the cover is operating correctly. If the exhaust system does not operate correctly, oil mist may cause the hinge of the small flip-up cover to malfunction, which may result in personal injury.

- **If you find any problems, please contact Agilent.**
- **Do not disconnect the exhaust hose from the foreline pump.**

WARNING

Toxic Gases

Immediately turn off the gas supplies at the source in the event the exhaust system is stopped for any reason. Refer to "Appendix I. Hydrogen Safety Guide" on page 168 and "Appendix J. Ammonia Safety Guide" on page 170. These, and other gases which may be used with the ICP-MS present a variety of risks.

CAUTION

Condensation

Ensure that the exhaust system is working correctly at all times. Instrument damage from condensation may occur due to reverse flow through the exhaust system.

IMPORTANT

Exhaust venting must comply with all local environmental and safety codes. Contact your Environmental Health & Safety (EHS) specialist.

Handling Solvents Correctly

WARNING

Organic and aqueous solvents can be hazardous if handled improperly. Unskilled, improper, or careless use of flammable solvents can create explosion hazards and fire hazards. This can result in death, or severe personal injury or burns. Always follow the precautions listed below to protect both operator and instrument:

- **Read the Material Safety Data Sheet (MSDS) for each solvent used.**
- **Prepare samples and transfer acids under a fume hood with adequate extraction.**
- **Wear gloves when handling acids or solvents.**
- **Wear safety glasses when handling any liquids.**
- **Cover volatile samples to minimize exposure to fumes and possible explosion hazard.**
- **Clean any spillage immediately using the approved laboratory procedures.**
- **Store and prepare samples away from the instrument to prevent corrosion.**
- **Use sample uptake tubing that is compatible with the sample solvent being pumped.**
- **Solvent spillage into instrument: please call the customer contact center.**

Acid digestion at atmospheric or increased pressure require these additional precautions:

- **Apply heat slowly, watching for a possible reaction after each increase in temperature.**
- **Drip a second acid or washing aid slowly into a hot sample while watching for signs of a vigorous reaction.**
- **Cool the digested sample before you transfer it, or dilute it slowly with water.**
- **Never use perchloric acid in a pressure digestion.**

WARNING

When using Organic solvents, extra caution should be observed. If the spray chamber drain is not pumped properly, the excess flammable solvents in the spray chamber may ignite and/or become explosive. If the torch injector clogs, the increased pressure in the spray chamber may cause the end cap to blow off exposing the solvent to ignition.

Checking the Drain Vessel

WARNING

The drain vessel contains the spray chamber effluent, which can be toxic. Improper handling of the vessel can result in a serious explosion or fire if incompatible substances accumulate. Corrosion of the vessel and connecting tube can result in leaks that may damage the instrument or cause bodily harm. If the effluent that is collected in the drain vessel contains toxic materials or solvents, follow approved laboratory procedures to safely dispose of this hazardous waste.

Make sure that the drain bottle is adequately ventilated (by the lab ventilation system, the same as for the ICP-MS mainframe) to deal with vapors from the bottle.

Clean the drain vessel every time you empty it by thoroughly flushing it with water. If it contained organic solvents, wash the drain vessel in acetone and let it dry.

Follow the procedures below to avoid exposure to the contents of the drain vessel:

- When using organic solvent, use a suitably sized waste container of appropriately resistant material to collect organic solvent.
- Put the drain vessel on the instrument table, where it is easy to check the liquid level.
- Tightly connect the drainage tubing from the spray chamber around the peristaltic pump to the drain vessel. Do not crimp the tubing.
- Check the drain vessel frequently. Empty it before you ignite the plasma.
- Be aware of the nature of the vessel contents. If the contents are toxic, dispose of them as hazardous waste. Also, always empty the vessel when you switch from aqueous to organic sample solutions.
- Check the tubing and vessel for deterioration. If the tubing becomes brittle or cracked, replace it. Organic solvents generally cause more rapid deterioration than aqueous solutions.

Compressed Gas Tanks

WARNING **Do not set the cell gas supply pressure higher than the specification of the ICP-MS instrument.**

WARNING **Compressed gas tanks must be handled with care. The contents of the cylinders also may be hazardous, depending on the gases you choose to use. All compressed gases (other than air) can create a hazard if they leak into the atmosphere. Even small leaks in gas supply systems can be dangerous. Any leak can create an explosion hazard, a fire hazard, or can result in an oxygen-deficient atmosphere. Such hazards can cause death, serious injury, asphyxiation, anesthetic effects, and serious damage to equipment and property.**

WARNING **Consult the cylinder, regulator and/or gas supplier for additional safety measures and ensure all staff are fully familiarized with all safety precautions.**

Storing Argon Safely

WARNING

Argon, which is used to create the plasma, is a dangerous gas only if it displaces the air you are breathing or is mishandled in its storage cylinder. Take the following precautions to prevent an explosion or suffocation hazard:

- Secure the cylinder valve caps and move the cylinder with an approved handcart.
- Gas cylinders release gas through a pressure-relief device under excessive heat. Always follow the gas supplier's recommendations for storage and use temperatures.
- Make sure that cylinders stored outside are not in direct sunlight or subjected to extreme temperatures. Place the cylinder on a level surface above ground.
- Attach the argon hoses tightly to both the instrument and gas source. Route the hoses so they cannot be damaged or crimped. Check for leaks using an electronic leak detector or pressure test.
- Make sure that there is adequate ventilation around the gas cylinders, especially if they are placed in a small storeroom.

Cell Gas

WARNING

Personal Injury Hazard

The exhaust gas tubing from foreline pump must be always connected to the rear port of the ICP-MS instrument to ensure that the cell gas evacuated by the foreline pump is exhausted via the external duct. Health hazards include the possibility of flammable, toxic or asphyxiating gas entering the room air.

WARNING

Health Hazard

Close all gas supply valves to the ICP-MS instrument before turning off the vacuum system. Health hazards include the possibility of flammable, toxic or asphyxiating gas accumulating in the vacuum chamber or instrument enclosure.

Individual stop valves must be installed for all gas lines for each individual ICP-MS instrument.

WARNING

Flammable gas (e.g. Hydrogen) and combustion-supporting gas (e.g. Oxygen) must always be placed in separate safety cabinets.

WARNING

Fully and strictly adhere to all local and national regulations and guidelines for the proper storage, handling and transport of all gases.

WARNING

3rd Cell Gas:

- Helium content of gas mix must be 80% or more.
- For example: NH3/He mix; the He content must be equal to or greater than 80%.
- Even if introducing a non-corrosive non-flammable gas to the cell, it must still be diluted with 80% or more Helium.

Oxygen Gas

WARNING

Oxygen has the following properties. Handle with care.

- Oxygen supports combustion of other materials. Materials which are noncombustible in air may become combustible in oxygen.
- When compared to being in air, the flammability range of materials increases in oxygen, and materials burn at lower temperatures.
- Open the oxygen cylinder valve slowly. Opening the valve too quickly can generate heat by adiabatic compression (a momentary high-temperature state caused by rapid compression of oxygen) and friction, increasing the risk of ignition.
- In high-density oxygen, there is a potential risk that materials such as metals (and metal powders), dust, and hydrocarbons (petroleum, greases, oils and fats, skin oils, and so on) can easily burn

WARNING

Smoking, open flames and other sources of ignition are prohibited in or near a facility which uses oxygen. Additionally, do not place inflammable and pyrophoric materials in the area. Make sure that all applicable local and national regulations and guidelines for the use and handling of oxygen are properly observed.

WARNING

Health hazards of oxygen:
The primary health hazard at atmospheric pressure is respiratory system irritation after exposure to high oxygen concentrations. Oxygen levels in air should be maintained above 19.5% and below 23.5%. Up to 50% oxygen can be breathed for more than 24 hours without adverse effects.
Prolonged exposure to high oxygen levels (>75%) can cause central nervous system depression: signs/symptoms can include headache, dizziness, drowsiness, poor coordination, slowed reaction time, slurred speech, giddiness and unconsciousness. In addition, note the following inhalation effects from acute exposure: may cause breathing difficulty; may cause coughing and chest pain; may cause lung damage; may cause soreness of the throat.

Other Gas

WARNING

- All Gases must be handled with care.
- Refer to "Appendix I. Hydrogen Safety Guide" on page 168 and "Appendix J. Ammonia Safety Guide" on page 170. Pay careful attention to safety management when using these gases.
- Refer to the MSDS (Material Safety Data Sheet) for all gases for safe handling information.
- Check for gas leaks periodically using the appropriate tools (leak detector, pressure test, etc.)

Allowing the Hot Instrument to Cool

WARNING

The torch and interface remain hot after the plasma is turned OFF. Do not touch the torch box or interface cones for 10 minutes after turning off the plasma to let them cool.

WARNING

Items in the vacuum chamber remain hot after the instrument shuts down. Do not touch them for 10 minutes after removing the chamber cover.

Nebulizer

WARNING

Personal Injury Hazard

Handle the nebulizer and other glassware with care, to avoid injury by broken glass.

Torch Box Cover

CAUTION

Always re-attach the torch box cover after maintenance. This cover reduces the emission noise from the plasma.

Peristaltic Pump

WARNING

Take care when you open and close the clamp of the peristaltic pump. Otherwise, you could get your finger caught between the stator and the rotator, or between the stator and the stopper.

Foreline Pump

WARNING

Foreline Pump oil is flammable. Keep away from fire.

WARNING

The pump oil may be very hot. Direct skin contact may cause burns.
If foreline pump oil is accidentally spilled on skin, in mouth, or eyes; wash immediately and thoroughly and seek expert medical attention.

WARNING

The surface of the Foreline Pump may be HOT, do not touch the pump until it has cooled.

Air Intake

CAUTION

Do not cover the holes for air intake on the ICP-MS.

Magnetic Emission

CAUTION

Electrical appliances installed within 5 cm of the far left side of the ICP-MS may not work correctly due to magnetic interference from the cool water inlet valve. Do not install electrical appliances on the left side of the ICP-MS.

Cautions for Toxic Materials

WARNING

There is a toxic hazard associated with disposal of components that contain beryllium or polyvinyl chloride (PVC.) Use caution to dispose of components that contain these materials.
The following components used in this instrument contain beryllium compounds (Table 1) or PVC (Table 2):

NOTE

Part numbers listed may be manufacturing part numbers (non-orderable; subject to change). As necessary, use the part description to identify components listed in tables 2 and 3 for disposal.

WARNING

Table 1. Parts that contain beryllium compounds

Agilent P/N	Description
G3280-40460	Feedthrough for Quad
G3280-60025	QPHV-1 Cable Assy
G3280-60026	QPHV-2 Cable Assy
G3280-60361	Power Sensor Assy for RF
G3660-60119	Q2 Mainfilter Cable Assy 1
G3660-60120	Q2 Mainfilter Cable Assy 2
G3660-60117	Q2 Prefilter Cable Assy 1
G3660-60118	Q2 Prefilter Cable Assy 2
G3660-60111	Q1 Prefilter Cable Assy 1
G3660-60112	Q1 Prefilter Cable Assy 2
G3660-60113	Q1 Mainfilter Cable Assy 1

Table 1. Parts that contain beryllium compounds (continued)

Agilent P/N	Description
G3660-60114	Q1 Mainfilter Cable Assy 2
G3660-60115	Q1 Postfilter Cable Assy 1
G3660-60116	Q1 Postfilter Cable Assy 2
G3660-60121	Q1 Entrance Cable Assy
G3666-60381	Hermetic Assy for Octopole Driver
G3660-60128	Hermetic Assy for Cell Temp CTRL
G3666-60383	Capacitor Assy EM gate 1, 2
G3666-60384	Capacitor Assy A-HV, G-HV
G3666-60385	Capacitor Assy P-Gate, P-HV
G8400-65800	EM Pulse Detector PCA
G3280-65811	RF Combiner-A PCA
G3280-65026	RF Power Module PCA
G3280-65035	QP RF PCA
G3666-65801	QQQ Cell Driver PCA
G8400-65815	Entrance PCA
G3280-80101	Opt Gas Flw Ctlr (80Ar/20O_2) w/Inlet Valve
G3666-80301	Option MFC 80%Ar/20%O2 1slmF.S. (For 8900 #100/#200)
G3280-80102	NH3 Mass Flow Controller
G3666-80320	4th Mass Flow Controller (Metal Seal)
G3660-60805	Cable: Main Q1
G3660-60818	QP Entrance Lens Cable Assy
G3660-60822	Cable Assy for 4th Cell Gas MFC
G3280-60213	Cable Assy for 3rd Cell Gas MFC
G8400-60455	EM Analog Signal Cable (To Detector Bd)

Table 2. Parts that contain PVC

Agilent P/N	Description
0890-2417	Vinyl Tube 5mm ID x 8mm OD
5042-0917	Sample Uptake Tubing for sample intro,12pk
5042-4709	Tube 6.35mm OD, 3.17mm ID (For sample drain)
5043-0015	Flared Sample Uptake tube, id 0.25
5064-8091	Motor Assy
5064-8099	5 Port Solenoid Valve
5182-7263	FAN 109R1224H141 (For quad tank coil cooling)
G1833-80217	HVB51X0340-FL-248390 (Interface Solenoid Valve)
G1833-80388	VINYL CASE 280 x 110 (Tool Kit Case)
G1833-80413	SPRING HOSE TE25x33xL1.5m (Vacuum hose between 8900 and MS40+)
G1833-80414	SPRING HOSE TE25x33xL3m (Extension vacuum hose between 8900 and MS40+)
G1833-80422	HOSE NCB12-10m, Braid hose (mainframe external) for cooling water
G1833-80423	HOSE NCB12-3m, Braid hose for MS40+ exhaust
G3280-60078	Cable: Intro PCA - Ar AMFC
G3280-60816	Cable: QP RF PCA to QP Controller PCA
G3280-60030	QP Tank Diagnostic Cable
G3280-60818	QP DC Cable Assy
G3280-60032	RF Power Sensor Cable Assy
G3280-60033	Unbalance Cable Assy
G3280-60034	RF Drive Power Cable Assy
G3280-60036	Cable: RF to MNB (for Phase Signal)
G3280-60037	XY Motor Cable Harness Assy
G3280-60038	Z Motor Cable Assy
G3280-60039	XY Start Pos Sensor Cable Assy
G3280-60040	Z Pos Sensor Cable Assy
G3280-60041	RF+48V Monitor Cable Assy
G3280-60047	Cable: Main Board to ORS Board

Table 2. Parts that contain PVC (continued)

Agilent P/N	Description
G3280-60048	Cable: Main Board to Quad Board
G3280-60049	Cable: Power Supply to Connection Board
G3280-60050	Cable: Sample Intro Brd to S/C Connector
G8400-60122	Cable Harness - Intro PCA to PP/3-Way SV
G8400-60454	Pulse Cable: Detector Board to Main Board
G3280-60055	Cable: (Flat) for HV Board Connection
G3280-60059	Power Cord w/IEC60309 plug and connector
G3280-60060	Power Cord (w/IEC60309 and NEMA L6-30P)
G8400-60453	Cable: Main Board to Connection Board, Ion Gauge
G8400-60451	Cable: Main Board to Connection Board, 1
G8400-60542	Cable: Main Board to Connection Board, 2
G3280-60065	Cable: HV Board A to HV Board B
G3280-60066	QP 48V Cable
G3280-60067	Thermo Sensor (For water manifold assy)
G3280-60205	Cbl: XYZ Init Sw/Small Cvr Sens-Intro Bd
G3280-60207	Cbl: Turbo Pump On/Off Sw-Connection Bd
G3280-60319	Fan assy for MNB
G3660-60806	Cable: CN BD to QQQ System IO BD
G3666-60211	Cable: Switch and Valve Harness
G3660-60802	Harness Cbl: XYZ Bd/RF Gen/Int Bd-Main Bd
G3280-60803	Cable: Octopole Driver Board Connection
G3660-60804	Cable: Plasma RF 48V Line
G3280-60805	Cover Switch Interlock
G3280-60811	Cable for Pirani Gauge
G3280-65026	RF Power Module PCA
G8400-65804	HV Board B (PCA)
G3666-80000	Power Supply (Main PSU)
G3666-80010	Power supply (Sub PSU)

Table 2. Parts that contain PVC (continued)

Agilent P/N	Description
G3280-80006	Motor for MNB
G3280-80300	Motor for XYZ
G3660-80101	Vaccum hose-TMP to TMP, TMP to IF/BK valve
G3660-80301	Vacuum Hose with ETFE for Interface
G3660-60800	Cable: Q1
G3660-60804	Cable: Plasma
G3660-60805	Cable: Main PCA to Q1 IF PCA
G3660-60806	Cable: CN PCA to QQQ Sys IO PCA
G3666-60211	Cable: Switch and valve harness
G3660-60802	Harness Cable, System
G3660-60821	PS Remote Cable
G3660-60812	Cable: 200VAC
G3666-60212	Cable: 4th MFC Shutoff Valve
G3666-60321	3 port air valve assy 4th MFC shutoff
G3660-60810	Cable: TMP2 to QQQ Sys IO PCA
G3660-60822	Cable Assy for 4th Cell Gas MFC
G3280-60213	Cable Assy for 3rd Cell Gas MFC
G8400-60460	MNB Cable

Moving the Agilent ICP-MS

CAUTION

Observe the precautions below when you move the ICP-MS.

- **Make sure that the Main Power Breaker (located on the rear) is turned OFF before you move the instrument.**
- **Make sure all the cables between other units are unplugged and the utility tubing is disconnected before you move the equipment (small adjustments can be made without unplugging and disconnecting the cables and tubing).**
- **When the instrument is installed or moved, check for leaks using an appropriate device.**
- **Before you move the instrument, secure the torch box using the shipping clamp.**

WARNING

The ICP-MS is heavy (about 143kg), so if you need to lift the instrument, it should be lifted by at least 4 people or with a mechanical lifter.

WARNING

If the exhaust fan is inoperable or there is insufficient air flow, do not start the plasma until the appropriate maintenance personnel have rectified the problem.

Do not cover the holes for air intake on the ICP-MS.

NOTE

The 8900 #100/#200 are equipped with the argon gas purifier (Big Universal Trap) which allows air to be exhausted easily by disconnecting the argon gas line from the source supply. To maintain the standard of the purifier, you need to prepare a new purifier after moving the ICP-MS instrument after disconnecting the argon gas connection.

Environmental Conditions, Compliance and Utility Requirements

Environmental Conditions

This equipment should be operated and stored under the temperature and humidity ranges described below:

Operation conditions:

- Temperature: 15 to 30 °C (59 to 86 °F)
- Relative humidity: 20 to 80 % (non-condensing)
- Altitude: Up to 2000 m
- Atmosphere: Non-Condensing; Non-Corrosive

For optimum instrument performance, the temperature variations should be less than 2°C in 1 hour.

Storage conditions:

- Ambient temperature: -20 to 60 °C
- Relative humidity: 15 to 80 % (non-condensing, non-corrosive)

This instrument is for indoor use only.

Installation category: II (See Note.)

Pollution degree: 2 (See Note.)

Equipment class: 1

NOTE

The “Installation category” refers to electrical cord requirements for surge voltage protection. It is also called the “Over voltage category”. “II” applies to electrical equipment.

“Pollution level” describes the degree to which a solid, liquid, or gas which deteriorates dielectric strength, is adhering. “2” applies to a normal indoor atmosphere.

This equipment requires the following space for ventilation, maintenance access, and easy access to the Main Power Breaker Switch. There must be a clear space of at least 60 cm (24 in) on all sides of the equipment. The bench in your laboratory must be able to support the entire ICP-MS system and other laboratory equipment.

Electromagnetic Compatibility

EN55011/CISPR11

Group 1 ISM equipment: group 1 contains all ISM equipment in which there is intentionally generated and/or used conductively-coupled radio-frequency energy which is necessary for the internal functioning of the equipment itself.

Class A equipment is equipment suitable for use in all establishments other than domestic and those directly connected to a low voltage power supply network which supplies buildings used for domestic purposes.

This device meets the requirements of CISPR11, Group 1, Class A as radiation professional equipment. Therefore, there may be potential difficulties in ensuring electromagnetic compatibility in other environments, due to conducted as well as radiated disturbances.

Operation is subject to the following two conditions:

1. This device may not cause harmful interference.
2. This device must accept any interference received, including interference that may cause undesired operation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, try one or more of the following measures:

1. Relocate the radio or antenna.
2. Move the device away from the radio or television.
3. Plug the device into a different electrical outlet, so that the device and the radio or television are on separate electrical circuits.
4. Ensure that all peripheral devices are also certified.

5 Ensure that appropriate cables are used to connect the device to peripheral equipment.

6 Consult your equipment dealer, Agilent Technologies, or an experienced technician for assistance.

Changes or modifications not expressly approved by Agilent Technologies could void your authority to operate the equipment.

EMC Declaration for South Korea

사용자안내문

This equipment has been evaluated for its suitability for use in a commercial environment.
When used in a domestic environment, there is a risk of radio interference.

이 기기는 업무용 환경에서 사용할 목적으로 적합성평가를 받은 기기로서 가정용 환 경에서 사용하는 경우 전파간섭의 우려가 있습니다.

ICES/NMB-001

This ISM device complies with Canadian ICES-001.

Cet appareil ISM est conforme à la norme NMB-001 du Canada.

Sound Emission Certification for Federal Republic of Germany

Sound pressure

Sound pressure Lp < 70 dB(A) according to DIN EN ISO 7779.

Schalldruckpegel

Schalldruckpegel LP < 70 dB(A) nach DIN EN ISO 7779.

Electrical Power (Voltage, Frequency, Amperage, Phase)

200-240 VAC, 50/60 Hz, 30 A, single phase

Supply voltage fluctuations do not exceed 10% of the nominal supply voltage.

Argon Gas Supply

Minimum Purity:	99.99 % (≥99.999 % recommended for ultra trace S/Si analysis)
Maximum Flow Rate:	20 L/min
Supply Pressure:	500 to 700 kPa

Cell Gas Supply

Gas:	Helium, Hydrogen, 3rd and 4th
Minimum Purity:	99.999 %
Maximum Flow Rate:	He 12 mL/min, H_2 10 mL/min, 3rd 10 mL/min, 4th 1.5 mL/min
Typical Pressure:	He 90-130 kPa (13-19 psi), H_2, 3rd and 4th 20-60 kPa (2.9-8.7 psi)

Cooling Water

Inlet Temperature:	15 to 40 °C
Inlet Pressure:	230 to 400 kPa
Minimum Flow Rate:	5.0 L/min

Exhaust Duct

Extraction Flow Rate:	5 to 7 m^3/min

Foreline Pump

Current Rating: 5 A

CAUTION

Do not use any foreline pump except those specified by Agilent for use with the Agilent ICP-MS.

NOTE

The rated current of foreline pumps (for use other than with the Agilent ICP-MS): 6.0 A for MS40+; 7.0 A for Dry pump (NeoDry36E), but the above listed rating of 5 A is applicable for use with the Agilent ICP-MS.